As is shown in §3.6, the grand partition functions F_i can be calculated from theory and spectroscopic data; because these functions are mass dependent k is mass dependent. In calculating the F_i's, all modes of energy must be included as well as the population of the different energy states.

For isotopes of the lighter elements, the activation energy term makes the main contribution to the reaction rate isotope effect, while for the heavier elements the vibrational frequency causing the decomposition into the products plays the larger role. Because the energy states usually are more separated for the isotopic molecules of the products and reactants than for the transition state, isotope effects are usually larger in reaction kinetics than in equilibria.

Studies of kinetic isotope effects are of considerable theoretical interest, particularly in organic chemistry. The practical applications are still meagre, but this will not necessarily be so in the future. An example is the decrease in metabolic rate for ^{13}C compounds, which has led to the suggestion of its use for treatment of certain diseases, as e.g. porphyria.

3.8. ISOTOPE SEPARATION PROCESSES

Many fields of fundamental science have found great advantage in using pure or enriched isotopes. The industrial use of nuclear power also requires the enrichment of particular isotopes, primarily of the uranium fuel but also separation of H, D and T in heavy water moderated reactors as well as nitrogen isotope separation for nitride fuels to be used in modern fast reactors (Ch. 21). The methods which have been developed to achieve isotopic fractionation may be divided into two groups.

(a) *Equilibrium processes* (§3.6). These processes consume little energy, but the size of the isotope effect in normal chemical equilibrium limits their use to the isotope fractionation of very light elements, usually atomic number less than about 10.

(b) *Rate processes* (§3.7). This includes processes which depend on such phenomena as ionic mobility, diffusion, electrolysis, electromagnetic separations, centrifugation, and kinetic processes of a chemical nature. While the isotopic effects in these processes are normally larger than for equilibrium processes, they require a large amount of energy and therefore have economic restrictions.

3.8.1. Multistage processes

Figure 3.8 shows a flow scheme for an isotopic fractionation process (or isotope *enrichment* process) based on an α-value near 1. Each *stage* consists of a number of *cells* coupled in parallel; in the Figure only one cell is shown for each separation stage, but in order to obtain a high product flow, the number of cells are usually high at the feed point and then decrease towards the product and waste stream ends. Each cell contains a physical arrangement, which leads to an isotope fractionation. Thus the atomic fraction

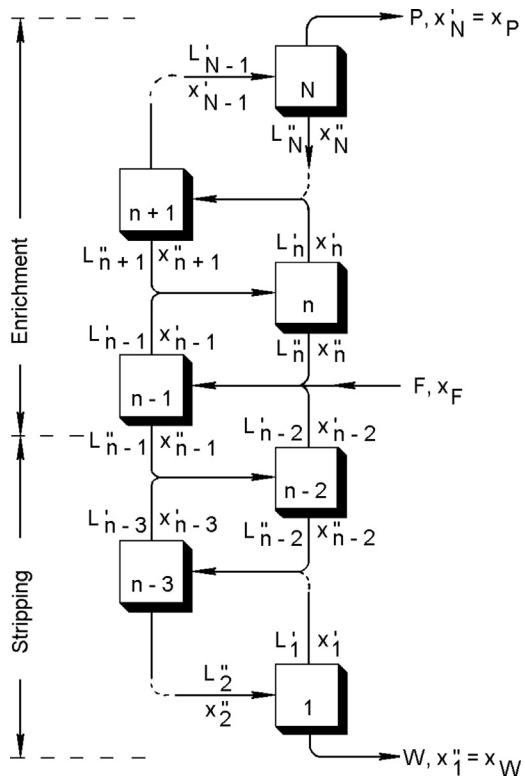


Figure 3.8 Flow arrangement for an ideal cascade with reflux for isotope separation.

for a particular isotope is different in the two outgoing streams from a cell; in the *product stream* the isotope is enriched (atomic fraction x'), while in the *waste stream* it is depleted (atomic fraction x''). The *separation factor* α is defined as the quotient between the isotopic ratios of the product and waste streams for a single step, thus ((3.3) and (3.4)):

$$\alpha = \zeta'/\zeta'' = [x'/(1-x')]/[x''/(1-x'')] \tag{3.61}$$

In most cases α has a value close to unity; $\alpha - 1$ is commonly called *enrichment factor.*

Since separation factors in general are small, it is necessary to use a multistage process to obtain a product with a high enrichment. The number of stages determines the degree of the enrichment of the product, while the number and size of cells in each stage determine the amount of product. This amount (P moles of composition x_P) is related to the amount of feed (F moles of composition x_F) and the amount of waste (W moles of composition x_W) by the equations

$$F = P + W \tag{3.62}$$

and

$$F x_F = P x_P + W x_W \tag{3.63}$$

From these equations we obtain

$$F = P\,(x_P - x_W)/(x_F - x_W) \tag{3.64}$$

$$W = P\,(x_P - x_F)/(x_F - x_W) \tag{3.65}$$

The number of stages required to separate feed into product and waste of specified composition is a minimum at *total reflux*, when $P = 0$. For this condition M. R. Fenske has derived a relation, which can be divided into one part for the enrichment:

$$N_P \ln \alpha = \ln[x_P(1 - x_F)/\{x_F(1 - x_P)\}] \tag{3.66}$$

and one for the stripping part of the cascade:

$$N_W \ln \alpha = \ln[x_F(1 - x_W)/\{x_W(1 - x_F)\}] \tag{3.67}$$

N_P and N_W are the minimum number of enrichment and stripping stages, respectively. In isotope separations α is often very close to one; ln α can then be replaced by $(\alpha - 1)$. In practice some product flow is desired, the fraction withdrawn at the enrichment stage being known as "*the cut*" P/F. The number of stages required to produce the composition x_P then increases. The most economic, and thus also the most common, type of cascade for processes with near unity α is the so-called *ideal cascade*. In this there is no mixing of streams of unequal concentrations, thus $x'_{n-1} = x''_{n+1}$ in Fig. 3.8. Although the number of stages required for a particular product is exactly twice the values given by (3.66) minus 1 and (3.67) minus 1,

$$N_{\text{ideal}} = 2N - 1 \tag{3.68}$$

the interstage flow becomes a minimum, thus minimizing the inventory and work required. Also for the ideal cascade, the minimum number of stages are obtained at total reflux.

The interstage flow rate L_i that must be maintained in order that there be any enrichment in the ideal cascade at the point, where the concentration x_i occurs, is given by

$$L_i = 2P(x_P - x_i)/[(\alpha - 1)x_i(1 - x_i)] \tag{3.69}$$

To obtain the flow rate in the stripping part, P is replaced by W, and $x_P - x_i$ by $x_i - x_W$.

For the enrichment of natural uranium (0.71% in ^{235}U) to 80% in ^{235}U, the minimum number of enrichment stages is found to be almost 3000 at an enrichment factor of 0.43%. The flow rate at the feed point becomes about 52 300 times the product flow P.

Isotope separation plants therefore become very large even for small outputs. Isotopic separation on a technical scale using cells with α near unity thus requires a large number of stages in order to obtain high enrichment and large amounts of materials in order to give a substantial amount of product. Small changes in α can have a large economic impact in isotopic separation.

3.8.2. Chemical exchange

As an example of an industrial isotope exchange process, let us consider the production of heavy water by the chemical reaction

$$H_2O(l) + HDS(g) = HDO(l) + H_2S(g) \qquad \begin{matrix} k = 2.32(32^\circ C) \\ k = 1.80(138^\circ C) \end{matrix} \tag{3.70}$$

From the values of the equilibrium constants k we see that the enrichment of deuterium in water increases with decreasing temperature. Use is made of this property in the two-temperature H_2O-H_2S exchange process known as the G-S process (Girdler-Sulphide), which is used in many countries to produce heavy water. A typical plant consists of several units as shown in Fig. 3.9. Through the upper distillation tower natural water flows downward and meets hydrogen sulfide gas streaming upwards. As a result of the exchange between H_2O and H_2S, heavy hydrogen is enriched in the water. In the lower tower, which is operated at a higher temperature, the equilibrium conditions are such that deuterium is enriched in the hydrogen sulfide and moves with that gas to the upper tower. No catalyzer is required in order to achieve rapid equilibrium in this reaction. The product of the process is water which is enriched in deuterium from the top tower and water which is depleted in deuterium from the bottom tower. The hydrogen sulfide circulates through both towers with no net loss.

Plants capable to produce a total of more than 1200 tons annually exsist in Canada, India and the US. The largest exchange towers are 60 m high and have a diameter of 6 m. In 5 units (only one unit is indicated in Fig. 3.8) the D_2O concentration is raised from 0.014% to about 15%. The final concentration to 99.97% D_2O is then usually made by distillation of water. The 1990 price for pure D_2O was $\sim$ US \$400 per kg. It is important to recognize in tracer applications that commercially available D_2O always contains some tritium, which is co-enriched with deuterium; 2–7 kBq kg^{-1} D_2O.

As another example, the lithium isotope ^7Li (used as ^7LiOH for pH control in some nuclear power plants because of its small neutron cross-section) is produced in 99.99% purity by counter current chemical exchange of lithium between an aqueous solution of LiOH and lithium amalgam. A separation factor of 1.06 to 1.07 is reported thus requiring about 260 theoretical stages to reach the desired purity with about 0.3% of ^7Li remaining in the depleted stream. Reflux of lithium is obtained at one end by electrolytic

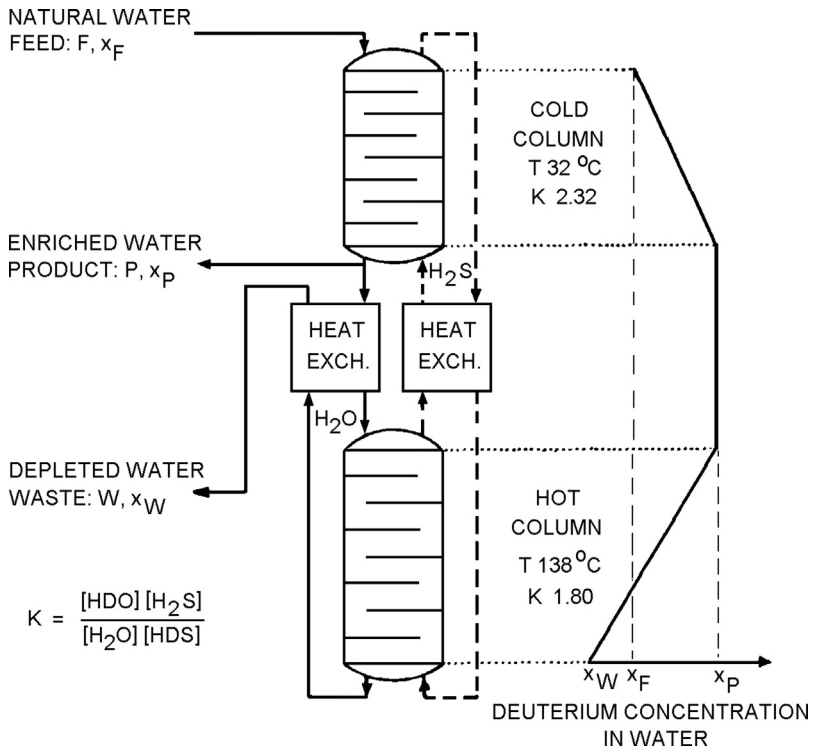


Figure 3.9 The water-hydrogen sulphide dual-temperature distillation process for enrichment of heavy water.

reduction of LiOH to Li(Hg) at a mercury cathode and at the other end by spontaneous oxidation of Li(Hg) on graphite by water producing hydrogen gas and LiOH.

3.8.3. Electrolysis

Electrolysis of water produces hydrogen gas at the cathode, which contains a lower proportion of deuterium than the original water. The isotope effect stems from the differences in the rates of dissociation of a proton (H^+) and a deuteron (D^+) from water, and the rates of neutralization of these hydrated ions, and thus has a kinetic basis. Depending on the physical conditions α-values between 3 and 10 are obtained. For $\alpha = 6$ it is necessary to electrolyze 2700 l natural water (deuterium content 0.014%) to produce 1 l water containing 10% deuterium, mainly as HDO. In a multistage process the hydrogen gas is either burnt to recover energy, or used in a chemical process, e.g. in ammonia synthesis by the Haber-Bosch process. Although this technique was once used industrially in Norway to produce ton amounts of pure D_2O, it is no longer considered economic except for final purification of D_2O.

3.8.4. Gaseous diffusion

In a gaseous sample the lighter molecules have a higher average velocity than the heavier molecules. In 1913 F. W. Aston in England showed that the lighter of the two neon isotopes ^{20}Ne, diffused through the walls of porous vessels somewhat faster than the heavier isotope, ^{22}Ne. In the gas the average kinetic energy of isotopic molecules must be the same, i.e. $1/2M_L v^2_L = 1/2M_H v^2_H$, where M_H and M_L are the masses of the molecules containing the heavy and light isotopes to be separated. The maximum theoretical separation factor in gaseous diffusion is given by

$$\alpha = v_L/v_H = (M_H/M_L)^{1/2} \tag{3.71}$$

The theory is more complicated, depending among other things upon the mean free path of the gaseous atoms, the pore length and diameter of the separating membrane, and the pressure difference over the membrane. If experimental conditions are carefully controlled, this theoretical maximum can be closely approached.

^{235}U is enriched through gaseous diffusion using the relatively volatile uranium compound UF_6. In addition to its volatility, UF_6 has the advantage that fluorine consists of only one isotope, ^{19}F. For the isotopic molecules $^{235}UF_6$ and $^{238}UF_6$, a value of 1.0043 is theoretically possible for α, cf. (3.71). The following conditions must be considered in the technical application of the separation.

(a) The cells are divided into two parts by a membrane which must have very small pores (e.g. 10–100 nm in diameter) in order to obtain isotopic separation. In order that large gas volumes can flow through the membrane, millions of pores are required for each square centimeter. Moreover, the membranes must have good mechanical stability to withstand the pressure difference across them.

(b) UF_6 sublimes at 64°C, which means that the separation process must be conducted at a temperature above this.

(c) UF_6 is highly corrosive and attacks most materials. The membrane must be inert to attack by UF_6. Water decomposes UF_6 according to the equation

$$UF_6(g) + 2H_2O(g) = UO_2F_2(s) + 4HF(g) \tag{3.72}$$

This reaction is quite undesirable as HF is highly corrosive and solid UO_2F_2 can plug the pores of the membranes. The tubing and cells of a plant are made principally of nickel and teflon to overcome the corrosion problems.

(d) In order to transport the large gas volumes and to keep the proper pressure drop across the membranes, a gaseous diffusion plant requires a large number of pumps. A large cooling capacity is needed to overcome the temperature rise caused by compression of the gas.

The work required to enrich uranium in ^{235}U increases rapidly with the mole fraction of ^{235}U in the product. Because of varying domestic prices on natural uranium, as well as

varying content of ^{235}U in uranium obtained from used reactor fuel elements, so-called *toll enrichment* has been introduced. In this case, the purchaser himself provides the uranium feed into the separation plant and pays for the *separative work* required to make his desired product out of the uranium feed provided. Separative work is defined as

$$\text{Separative work} = W\ V(x_W) + P\ V(x_P) - F\ V(x_F) \tag{3.73}$$

where the *separation potential* $V(x_i)$ is defined by

$$V(x_i) = (2x_i - 1)\ \ln\{x_i/(1 - x_i)\} \tag{3.74}$$

As seen from (3.73), separative work has the dimension of mass, and can be thought of as the mass flow rate multiplied by the time required to yield a given quantity of product. The cost of isotope separation is obtained by assigning a value to one separative work mass unit (kgSW or SWU). A 1 GWe nuclear light water reactor station requires about 180×10^3 SWU in initial fueling and then $70–90 \times 10^3$ SWU for an annual reload.

In §3.8.1 the number of stages and the interstage flow relative to the product flow was given for enrichment of ^{235}U from its natural isotopic abundance of 0.71% to a value of 80%. With a waste flow in which the isotopic abundance of ^{235}U is 0.2%, (3.64) and (3.65) shows that for each mole of product obtained 156 moles of feed are necessary. In more recent designs the concentration of ^{235}U in the waste is increased to ~0.3% to minimize cost. Isotope separation through gaseous diffusion is a very energy-consuming process due to the compression work and cooling required. An annual production of 10 MSWU requires an installed capacity of ~2900 MW in present plants, or ~2500 kWh SWU^{-1}. Improved technology may reduce this somewhat. Gaseous diffusion plants are known to exist in Argentina, China, France, Russia and the United States. The combined capacity of these plants was about 40 MSWU/y at the end of 2000.

3.8.5. Electromagnetic isotope separation

During the Manhattan Project of the United States, electromagnetic separation was used to obtain pure ^{235}U. This process is identical in theory to that described for the mass spectrometer. The giant electromagnetic separators were called *calutrons* (California University Cyclotrons) and were after World-War II used at Oak Ridge National Laboratory to produce gram amounts of stable isotopes of most elements up to a purity of 99.9% or more. Large capacity electromagnetic separators have also been developed and operated in the former Soviet Union and a few other countries. Gram amounts of electromagnetically separated stable isotopes of most elements are commercially available on the world market in high isotopic purity. Electromagnetic separators are also used for on-line separation of nuclear reaction products (Ch. 17). A large number of such separators were also secretly built and operated by Irak for U-enrichment before the Gulf war.

3.8.6. Gas centrifugation

Though gas diffusion has been the dominating process for ^{235}U enrichment, its small separation factor, large energy consumption, and secrecy about the technique have encouraged interest in finding other and more advantageous processes. Most effort has been put into developing centrifugal processes because of the large separation factors achievable.

In a gas centrifuge, light molecules are enriched at the center and heavy molecules at the periphery (Fig. 3.10). It can be shown that the ratio of radial concentration to the axial concentration for isotopic molecules under equilibrium conditions and at low pressure (avoiding remixing due to the Brownian movement) is given approximately by

$$x_P/x_F \approx e^{\delta n} \tag{3.75}$$

for n centrifuges (stages) connected in series. Here

$$\delta = (M_H - M_L)10^{-3}\nu_r^2\, r^2/2\mathbf{R}T \tag{3.76}$$

where r is the centrifuge radius and ν_r the speed of rotation. The separation factor $\alpha \approx 1 + \delta$.

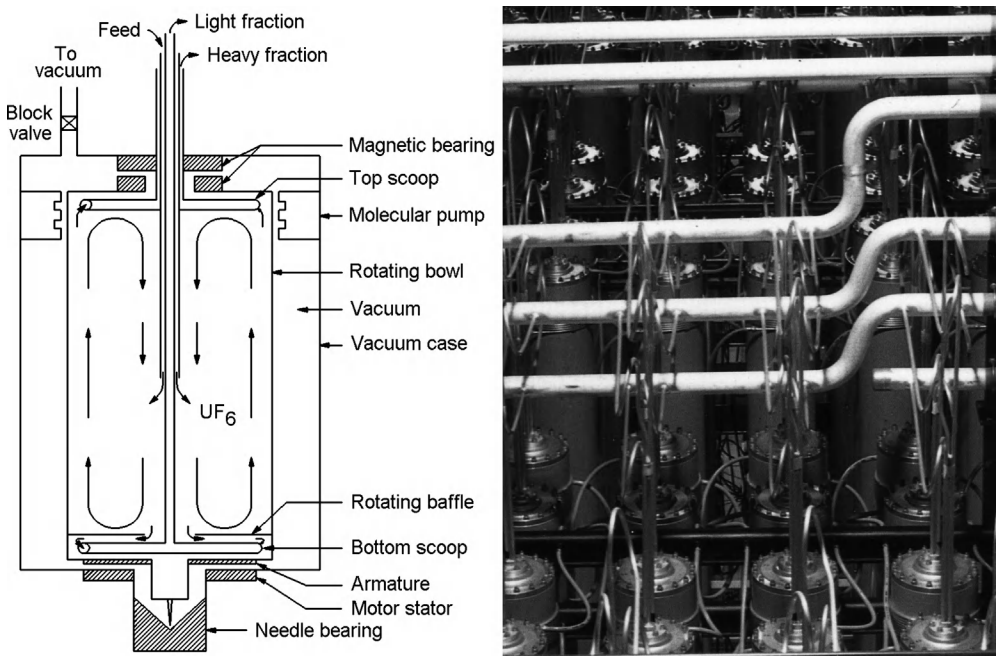


Figure 3.10 Gas centrifuge for $^{235}UF_6$ enrichment. Right shows detail of centrifuge cascade in operation at Almelo.

Estimated values for present centrifuges operating on UF_6 are: length 1−2 m; radius 0.2−0.4 m; rotational speeds of $50-80 \times 10^3$ rpm. The enrichment obtainable in one stage is limited by the material strength of the centrifuge bowl; the present materials limit $v_r\, r$ to < 800 m s^{-1}. Typical separation factors are 1.4−3.9 per stage, thus about 10 stages are required to enrich ^{235}U from 0.72% to 3% with a 0.2% tail. As compared, a diffusion plant would require ~1300 stages. The corresponding figures for production of 80% ^{235}U are ~45 and ~3600, respectively. Though rather few stages are required to upgrade natural uranium to reactor quality, a very large number of centrifuges are needed to produce large quantities of enriched material or highly enriched material.

Current centrifuge technology requires ~3% of the power consumed by a diffusion plant, or 50−100 kWh SWU^{-1}. This makes their environmental impact minimal, as compared to gas diffusion plants, which require substantial electric power installation and cooling towers with large water vapour effluent. Smaller plants of a few MSWU/y are economical, and their output can be readily multiplied by installation of parallel processing lines. The very large number of centrifuges required due to their small size (each having a capacity of = 15 SWU/y) and limited life-time does not lead to excessive construction costs due to continuous mass production on-site. On the whole, centrifuge separation now seems to have a lower enrichment cost than large scale diffusion plants. Centrifuge performance has been increased by a factor of ~25 since 1980 and is predicted to improve further.

Many plants are now operating, the largest known being near Ekaterinburg (Russia, 9 MSWU/y; Russian centrifuge plants have total capacity of ~19 MSWU/y), Almelo (the Netherlands, 1.7 MSWU/y), Capenhurst (United Kingdom, 1.1 MSWU/y) and Gronau (Germany, 1.1 MSWU/y).

3.8.7. Other methods of isotope separation

In theory all physicochemical procedures are capable of isotope separation. Some other methods which have been studied include distillation, solvent extraction, ion exchange, photoionization and photoexitation.

Tons of D_2O are purified annually in India by cryogenic *distillation* of hydrogen. Tenths of kilograms of pure ^{13}C and ^{15}N have been produced at the Los Alamos Scientific Laboratory through distillation of NO and CO at liquid air temperature. At the same time a fractionation between ^{16}O and ^{18}O occurs.

A *continuous ion-exchange* isotope separation process for uranium enrichment has been developed in Japan. Few details of this process have been disclosed. However, it is known that "reflux" is obtained by oxidation and reduction of U^{4+} and UO_2^{2+}. A demonstration plant with a capacity of 2 kSWU/y has been in operation at Hyuga.

A method of separation, involving passage of a mixture of UF_6 and helium or hydrogen at very high velocities through a *nozzle*, as seen in Fig. 3.11, has been

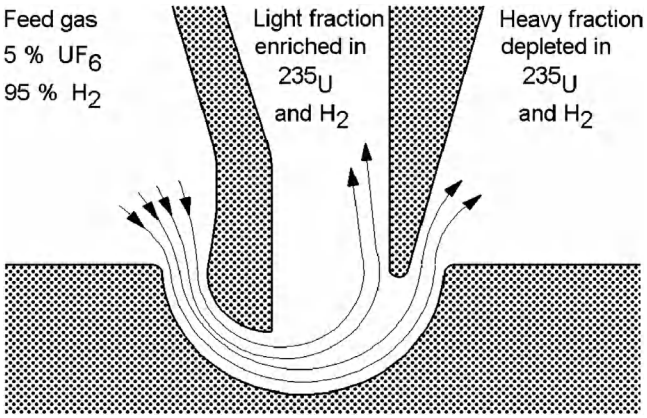


Figure 3.11 Section through a separation nozzle arrangement showing stream lines.

developed by E. W. Becker in Germany and in South Africa. The technique is sometimes referred to as "static" or "stationary-walled" gas centrifugation. The separation factor is typically 1.01 - 1.03 per stage, *i.e.* about three times better than in the gaseous diffusion process, and offers great possibilities for further improvements. Thus while the diffusion process requires about 1200 stages for a 3% ^{235}U enrichment (with 0.2% tails), the nozzle technique will require only about 500 stages. However, the power consumption is said to be larger, $\sim$3000 kWh SWU^{-1} at a capacity of 10 MSWU/y, as compared to $\sim$2500 kWh SWU^{-1} for a diffusion plant of the same size. The operating cost is said to be higher than for both the diffusion and the centrifuge methods. A plant based on the original German design is being built in Brazil (10 kSWU/y). A 0.3 MSWU/y separation plant at Valindaba, South Africa, used its own version of the nozzle technique (Helikon process), but is now shut down.

A number of *photoionization* and *photoexcitation* processes are being investigated for isotopic separation, especially of uranium. In one such process UF_6 is irradiated by a laser beam, producing selective vibrational excitation in the $^{235}UF_6$ molecule (cf. §3.5). By irradiation with ultraviolet light (possibly, but not necessarily, by laser) the excited molecule is caused to dissociate, leaving $^{238}UF_6$ undissociated. It is important that the ultraviolet pulse follows quite rapidly after the laser pulse, so that the vibrationally excited $^{235}UF_6$ molecule does not lose its excitation energy through collision with surrounding molecules. It is obvious that this necessitates gas phase reactions. The $^{235}UF_n^{6-n}$ ion formed through the dissociation ($n < 6$) is then collected by the action of electromagnetic fields. This technique is not limited to UF_6; pure uranium metal vapor and plutonium compounds have been separated into their isotopic constituents by two or three photon ionization with laser light (e.g. the US AVLIS process and the French SILVA process). Another alternative is to selectively excite $^{235}UF_6$ molecules by laser light in the presence of a reactive gas (the CRISLA process). The excited molecules then reacts preferentially with the gas

forming molecules with lower vapor pressure than UF_6. Although research in these areas has indicated a large scale feasibility of several similar processes, no predictions can yet be made of their technological value. The energy consumption for the quantum processes discussed above are in the range 10–40 kWh SWU^{-1}. However, economic estimates indicate that their enrichment cost falls in the same price range as for centrifuge based plants. Hence, the interest in these methods has decreased and development of the ALVIS and SILVA processes has been terminated. Finally, it should be pointed out that the general concept of separative work is not valid for very high separation factors leading to very few units in a cascade (and always for mixtures of more than two isotopes).

3.9. EXERCISES

3.1. How many atoms of ^{235}U exist in 1 kg of uranium oxide, U_3O_8, made of natural uranium?

3.2. What is the atomic fraction of deuterium in water with the mole fraction of 0.81 for H_2O, 0.18 for HDO, and 0.01 for D_2O?

3.3. The translational energy of one mole of gas is given by $3/2\mathbf{R}T$, which corresponds to an average thermal molecular velocity v (the root mean square velocity), while the most probable velocity $v' = \sqrt{0.67}\, v$.

(a) What is the most probable velocity of a helium atom at 800°C?

(b) What voltage would be required to accelerate an α-particle to the same velocity?

3.4. A Nier type mass spectrometer has a fixed radius of curvature of 5 cm and a magnetic field of 3000 G; 1 G = 10^{-4} T. At what accelerating voltage will a Na^+ ion be brought to focus at the ion collector?

3.5. In a Dempster type (constant B and V) mass spectrograph utilizing 180° degree focusing, the ions $^{12}C^+$ and $^{11}BH^+$ are recorded simultaneously, the latter ion having a slightly larger orbit diameter. The separation between the lines recorded on the photographic plate is 0.0143 cm and the orbit diameter for the $^{12}C^+$ ion is 20 cm. What is the atomic mass of ^{11}B?

3.6. When the rotational quantum number n_r goes from 0 to 1 in $H^{35}Cl$, it is accompanied by the absorption of light with a wave number of 20.6 cm^{-1}. From this it is possible to calculate the interatomic distance between hydrogen and chlorine in the molecules. What is this distance?

3.7. In one mole of a gas at STP (standard temperature and pressure, i.e. 0°C and 1 atm) a small fraction of the molecules have a kinetic energy = $15\mathbf{k}T$. (a) How many such molecules are there, and (b) what would their temperature be if they could be isolated?